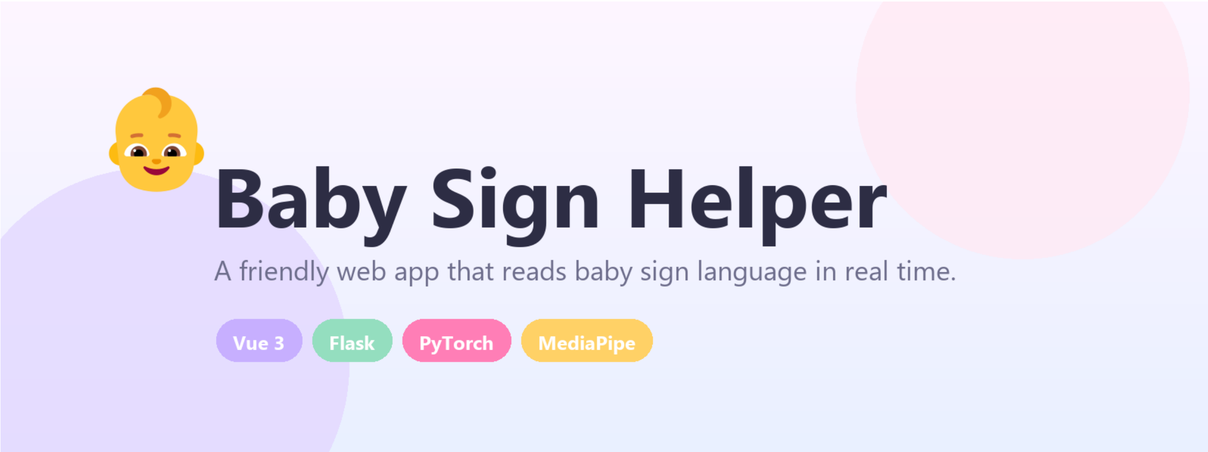

Baby Sign Helper

System Design Document

Version 1.0
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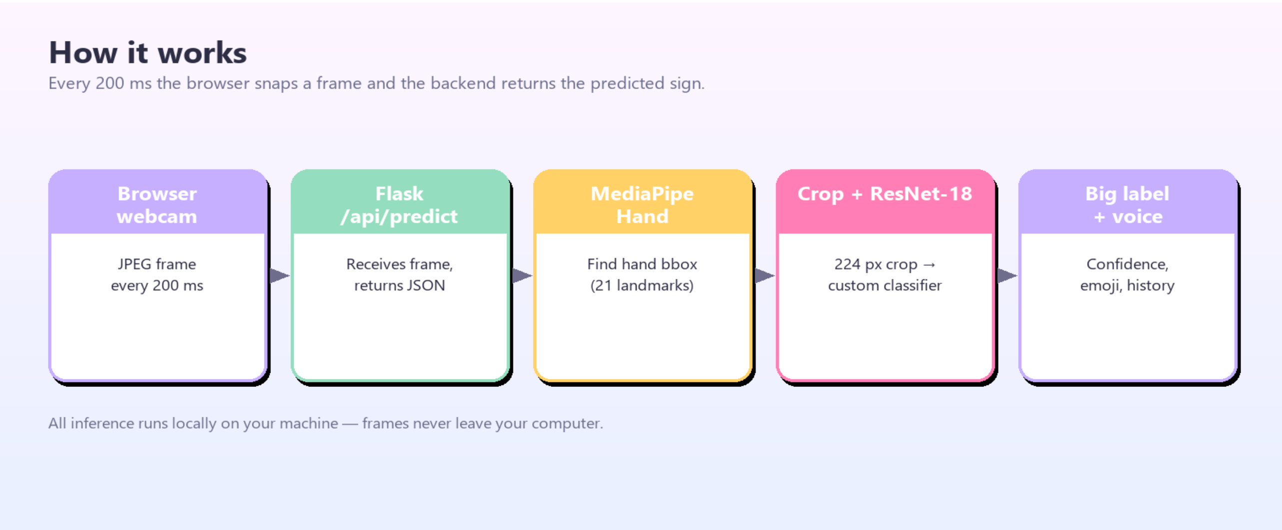
Owner: RahmanBhuiyan
github.com/RahmanBhuiyan/baby-sign-language-webapp

Built on top of RoxyDiya/BABY-SIGN-LANGUAGE-RECOGNITION (Sapienza University of Rome)

1. Overview

Baby Sign Helper is a friendly cross-platform application that reads baby sign-language gestures in real time and translates them into spoken words. A Vue 3 front-end captures webcam frames every 200 ms and POSTs them to a Flask backend hosted on Hugging Face Spaces; the backend uses Google's MediaPipe HandLandmarker to find the hand, then feeds the cropped region into a fine-tuned ResNet-18 classifier (12 classes) and returns the top-1 label plus confidence. The same Vue app is packaged with Capacitor into an Android APK so parents can install it on a phone and use the app on cellular data.

2. Architecture



Every 200 ms the browser snaps a frame from the device camera, encodes it as a base64 JPEG, and POSTs it to /api/predict. The Flask backend decodes the image, runs MediaPipe HandLandmarker to find the hand bounding box, crops with a 20 px offset, resizes to 224x224, normalizes for ImageNet, and runs the ResNet-18 classifier. The top-1 label and confidence return as JSON. The frontend shows a big emoji + word card, speaks the word through the Web Speech API when confidence is high, and keeps a rolling history of the last five signs.

3. Components

Component	Role	Technology	Runtime location
Web frontend	Camera capture + UI rendering, sends JPEG frames to /api/predict, plays back the spoken word	Vue 3 (Composition API) + Vite	Browser / Android WebView
Flask API	Decodes JPEG, calls hand detector, crops 224x224, runs ResNet, returns top-1 label	Flask 3 + flask-cors + gunicorn	Docker container on HF Spaces
Hand detector	Locates the hand bbox using 21 landmarks	MediaPipe Tasks HandLandmarker	Same container as Flask
Classifier model	ResNet-18 backbone with custom 256-unit dropout head over 12 classes	PyTorch 2.x + torchvision	Same container as Flask
Android shell	Wraps the built Vue app as a native APK; handles getUserMedia permission	Capacitor 7 + Android WebView	User's Android phone
CI/CD	Builds the Vue app, generates the Android project, signs the debug APK	GitHub Actions (Node 20, JDK 21, Android SDK)	GitHub-hosted Ubuntu runner
Source control	Two remotes from one working copy	Git	GitHub (code + workflow) and HF Spaces (Docker deploy)

4. Tech stack

Layer	Technology	Version	Purpose
Frontend framework	Vue	3.5.x	Reactive component model + Composition API
Frontend bundler	Vite	5.4.x	Dev server + production build
Mobile shell	Capacitor	7.x	Wraps the Vue app as Android APK
Browser APIs	getUserMedia, Web Speech API	native	Camera capture + voice playback
Backend framework	Flask	3.x	HTTP routing
WSGI server	gunicorn	22.x	Production process manager
CORS	flask-cors	6.x	Allow APK / browser cross-origin calls
ML framework	PyTorch	2.x	Inference
Computer vision	OpenCV	4.13.x	Image decoding + colour conversion
Hand detection	MediaPipe Tasks	0.10.x	21-landmark hand keypoint model
Image preprocessing	Pillow	12.x	JPEG decode
Numerics	NumPy	2.x	Tensor manipulation
HTTP client (auto-download)	requests	2.x	Fetch model weights on first boot
Container base	python:3.11-slim (Debian 12)	n/a	Backend runtime image
Hosting (backend)	Hugging Face Spaces (Docker SDK, free CPU)	n/a	Public HTTPS endpoint
Hosting (frontend)	Local Vite dev / Android APK / GitHub Pages (optional)	n/a	User-facing surface
CI runner	GitHub Actions ubuntu-latest	n/a	APK build pipeline

5. API specification

Method	Path	Request body	Response	Notes
GET	/api/health		{"status":"ok"}	Liveness probe; used by HF Space healthcheck and the APK on startup
POST	/api/predict	{"image":"data:image/jpeg;base64,..."}	{"has_hand":bool, "label":str|null, "confidence":float}	Top-1 prediction. has_hand=false means MediaPipe could not find a hand; label is null in that case.

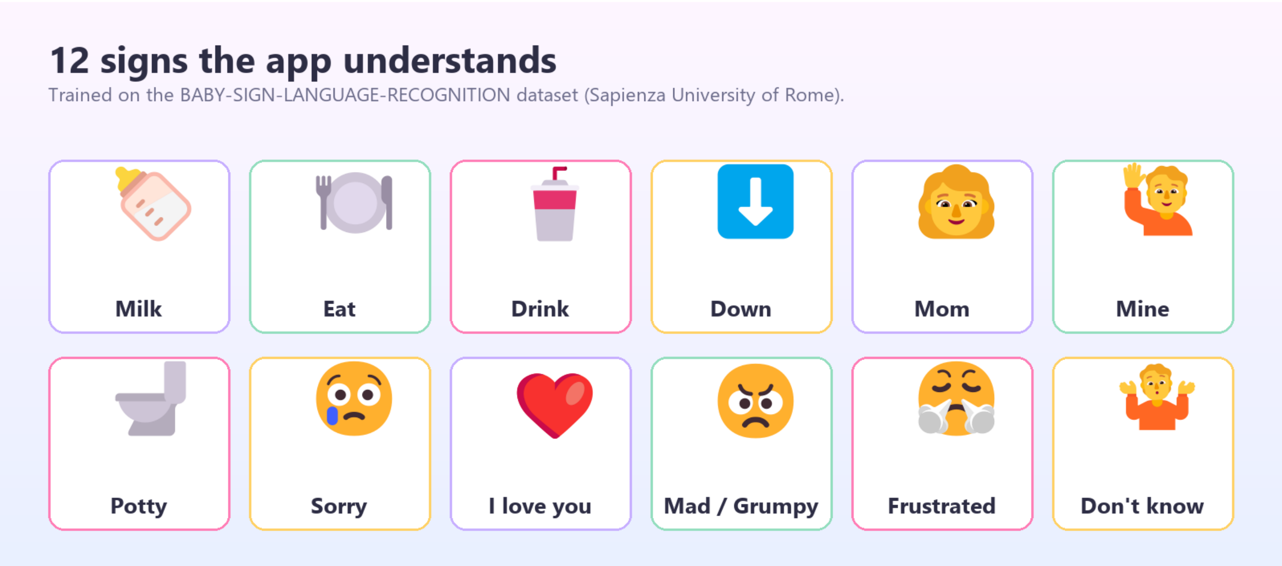
6. Data flow (per inference)

Step	Actor	Action	Payload / shape
1	Browser / APK	Capture frame via getUserMedia + canvas.toDataURL	base64 JPEG, 640x480
2	Browser / APK	POST to /api/predict	{image: ...}
3	Flask API	Decode base64 -> Pillow -> NumPy BGR	ndarray HxWx3
4	Hand detector	Run MediaPipe HandLandmarker -> 21 keypoints -> bbox	(x, y, w, h)
5	Flask API	Crop bbox + 20 px offset, resize 224x224, ImageNet-normalize	tensor 1x3x224x224
6	Classifier	ResNet-18 -> custom head -> softmax	12-vector probabilities
7	Flask API	argmax + lookup label name + round confidence	JSON {has_hand, label, confidence}
8	Browser / APK	Render big emoji + word + confidence bar; speak word if >=80%	DOM update + speechSynthesis.speak
9	Browser / APK	If confidence >=80%, append to recent-signs history (max 5, 3 s dedupe)	UI strip update

7. ML model details

Property	Value
Architecture	ResNet-18 (ImageNet-pretrained backbone) + custom classifier head
Custom head	Linear(512 -> 256) -> ReLU -> Dropout(0.5) -> Linear(256 -> 12)
Parameters	~11.31 M total
Input shape	3 x 224 x 224, ImageNet normalize (mean=[0.485,0.456,0.406], std=[0.229,0.224,0.225])
Output	12-class softmax (see Recognized Signs sheet)
Source	RoxyDiya/BABY-SIGN-LANGUAGE-RECOGNITION (Sapienza University of Rome)
Weights file	signlanguage_model.pth (44 MB), auto-downloaded on container boot
Hand detector weights	hand_landmarker.task (8 MB) from Google's MediaPipe model zoo
Inference cost	~150 ms per frame on HF Spaces free CPU

8. Recognized signs



9. Confidence rules

Condition	UI behaviour
>= 80%	Green bar; word is spoken aloud; chip is added to recent-signs history
50-79%	Yellow bar; label shown but not spoken; not added to history
< 50%	Predictions hidden; UI shows 'Show me a sign'
No hand detected	UI shows 'Looking for hands...'; backend returns has_hand=false
Repeat suppression (speech)	Same word not spoken twice within 2 s
Repeat suppression (history)	Same word not added to history twice within 3 s

10. Deployment topology

Environment	Service	URL / path	Notes
Production backend	Hugging Face Space (Docker)	https://codercarry-baby-sign-language-webapp.hf.space	Public HTTPS, free tier
Production backend (alt)	Page	https://huggingface.co/spaces/CoderCarry/baby-sign-language-webapp	Build logs + git remote
Source repo	GitHub	https://github.com/RahmanBhuiyan/baby-sign-language-webapp	Code + GitHub Actions workflows
APK artifact	GitHub Actions	Actions tab -> Build Android APK run -> Artifacts	Debug-signed, ~4 MB, regenerated on push
Local dev frontend	Vite dev server	http://localhost:5173	npm run dev in frontend/
Local dev backend	Flask dev server	http://localhost:5000	py -3.13 backend/app.py
CI variable	VITE_API_BASE	set in GitHub repo Settings -> Actions -> Variables	Baked into APK at build time

11. File manifest (key paths)

Path	Purpose
backend/app.py	Flask routes: /api/health, /api/predict, /(SPA fallback)
backend/model_loader.py	Singleton ResNet18CustomClassifier + auto-download .pth
backend/hand_detector.py	MediaPipe Tasks HandLandmarker singleton + auto-download .task
backend/inference.py	predict(b64_jpeg) -> {has_hand, label, confidence}
backend/requirements.txt	Python deps incl. gunicorn
backend/Dockerfile	HF Spaces image: non-root user, libgles2, gunicorn
backend/.dockerignore	Excludes __pycache__ and models/ from image
backend/README.md	HF Spaces frontmatter (sdk: docker, app_port: 7860)
frontend/src/App.vue	Top-level layout, wires composables to components
frontend/src/components/CameraView.vue	getUserMedia + 200 ms capture loop
frontend/src/components/LabelCard.vue	Big emoji + label + colored confidence bar
frontend/src/components/HistoryStrip.vue	Last 5 signs with relative time
frontend/src/components/SpeakToggle.vue	Voice on/off pill
frontend/src/composables/usePredictor.js	POST to /api/predict with in-flight guard
frontend/src/composables/useSpeech.js	Web Speech API + 2 s repeat-suppress
frontend/src/composables/useHistory.js	Ring buffer of 5, 3 s dedupe
frontend/src/constants/labels.js	Label -> emoji + pretty name mapping; HIGH_CONF=0.80
frontend/vite.config.js	Vite + /api proxy to :5000
frontend/capacitor.config.json	appId: com.rahmanbhuiyan.babysignhelper, webDir: dist
frontend/package.json	npm scripts + Capacitor deps
.github/workflows/android.yml	Builds APK on every push + workflow_dispatch
docs/*.png	README slide images (hero, architecture, features, signs grid)
system-design/generate.py	This script - produces the PDF and Excel

12. Future work

Item	Description
Serve frontend from HF too	Multi-stage Dockerfile builds the Vue app and Flask serves it, so codercarry-baby-sign-language-webapp.hf.space/ shows the UI instead of the placeholder 503.
On-device TFLite	Convert ResNet-18 to TFLite; run inference inside the APK so the app works offline and feels instant.
Play Store release	Generate keystore, build signed AAB in CI, write privacy policy, fill Play Console listing.
Persistent paid backend	Move HF free-tier backend to Render/Fly (~$5/mo) to remove cold-start delay.
More signs	Retrain with additional baby-sign-language classes (e.g. 'more', 'all done', 'thank you').
Authentication / rate limiting	Add an API key check on /api/predict if the app gets shared publicly.